

LECTURE L7-8 • MODULE 2 • CO2

Univariate EDA

Week 4 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 3-10 | Laboratory: P7-8

TOPICS COVERED

- Numerical variables: boxplot, density, distribution comparison
- Categorical variables: frequency tables, bar plots
- Detecting skewness and anomalies

Two families, used in order

Non-graphical technique produces numerical summaries — quantiles, moments, frequency and contingency tables, correlation matrices. It is precise, reproducible and easy to tabulate, and it is blind to shape. Anscombe's quartet is four datasets sharing means, variances and correlation coefficients while looking completely different.

Graphical technique reveals shape, gaps, clusters and outliers instantly, and is sensitive to choices you make — bin width, axis limits, smoothing bandwidth.

Use both, in this order: compute the summary, plot it, then reconcile the two when they disagree.

The four questions

Answer all four for every column before looking at a single relationship.

1. **What is the centre?** Mean or median, whichever the skew justifies. State which you used.
2. **What is the spread?** Standard deviation or IQR, plus the range — which exposes impossible values.
3. **What is the shape?** Symmetric, skewed, bimodal, uniform, zero-inflated. This decides everything downstream.
4. **What is unusual?** Outliers, gaps, spikes at round numbers, impossible values.

Choosing a distribution plot

PLOT	SHOWS	WEAKNESS
Histogram	frequency per bin	bin width changes the story
Density (KDE)	smoothed shape	bandwidth invents smoothness; spills past data limits
Boxplot	five-number summary	hides multimodality completely
Violin	density plus summary	cluttered with many groups
ECDF	cumulative proportion	less intuitive to a general audience

Default to histogram plus boxplot. Add a density overlay when comparing groups; reach for the ECDF when bin width is contested, since it requires no binning choice at all.

Reading anomalies off a plot

- A **spike at exactly zero** usually means "not recorded", not "measured as zero".
- **Two humps** mean two populations are mixed — find the variable that separates them.
- A **hard wall** at a round number suggests a cap, truncation or data-entry limit.
- **Spikes at multiples of 5 or 10** indicate self-reported or rounded data.
- A **long thin tail** is skew; **isolated far points** are candidate outliers. Different problems.

In `pima.csv` this reading finds 374 zeros in `Insulin` and 227 in `SkinThickness` — every one a disguised missing value that `isna()` cannot see. Module 3 treats them; Module 2 finds them.

Categorical univariate

Frequency tables with counts *and* proportions. Check cardinality (5 levels is a bar plot, 500 is a grouping problem) and imbalance. The shoppers `Revenue` flag is roughly 84.5% / 15.5%; the Pima outcome is 65% / 35%. Record this now — imbalance discovered here changes your sampling and evaluation strategy in Module 4, where a model predicting "never" would score 84.5% accuracy.

KEY TERMS

Non-graphical EDA	Graphical EDA	Anscombe's quartet	Kernel density estimate	ECDF
Zero-inflation	Multimodality	Class imbalance		

COMMON ERRORS TO AVOID

- Presenting a single histogram without checking a second bin width
- Trusting a KDE that extends past the physical limits of the variable
- Using a boxplot alone on data that turns out to be bimodal
- Failing to record class imbalance during univariate analysis

READINGS — ALL FREE TO ACCESS

1. seaborn — visualising distributions
<https://seaborn.pydata.org/tutorial/distributions.html>
2. NIST/SEMATECH — graphical techniques
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
3. Berkeley Data 100 — EDA notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. Two variables share a mean and standard deviation. Name two plots that would distinguish them.
2. A rate variable's KDE shows density below zero. What went wrong?
3. Why is a spike at exactly zero worth more attention than a point in the far tail?
4. At 15% positive class, what accuracy does a model that never predicts positive achieve?